

STARAYA RUSSA, Russian Federation

WMO: 262750

Lat: 58.017N

Lon: 31.317E

Elev: 82

StdP: 14.65

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.3	-6.1	-16.4	2.3	-12.0	-9.9	3.2	-5.7	17.5	30.7	15.9	31.4	2.3	340	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	83.6	69.2	80.1	67.3	76.9	65.4	71.0	80.6	69.0	77.5	67.1	74.7	5.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.6	101.8	75.9	65.8	95.4	73.8	64.0	89.5	71.5	34.8	80.7	33.2	77.3	31.6	74.9	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	13.0	11.5	DB	-17.1	87.9	7.9	3.5	-22.9	90.4	-27.5	92.4	-32.0	94.4	-37.7	96.9	(4)
(5)			WB	-17.4	73.4	7.9	2.9	-23.1	75.5	-27.7	77.2	-32.1	78.8	-37.8	80.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	21.0	20.9	29.8	43.3	54.0	60.6	65.7	62.3	53.2	42.3	32.3	25.7
	DBStd	18.06	12.71	13.14	8.78	7.85	7.82	6.44	5.78	5.58	6.49	7.40	9.10	11.03
	HDD50	4206	900	816	627	230	49	3	0	1	41	256	531	752
	HDD65	8296	1365	1236	1092	651	350	162	61	120	356	705	981	1217
	CDD50	1546	0	0	0	29	172	322	487	382	137	17	0	0
	CDD65	161	0	0	0	0	8	31	83	37	2	0	0	0
	CDH74	1370	0	0	0	4	95	268	679	310	14	0	0	0
	CDH80	302	0	0	0	0	16	56	168	62	0	0	0	0
Wind	WSAvg	5.4	6.1	5.9	5.9	5.6	5.2	4.7	4.0	4.3	4.5	5.4	6.2	6.6
Precipitation	PrecAvg	25.7	1.8	1.4	1.3	1.5	2.4	2.8	3.0	3.0	2.2	2.4	2.2	1.7
	PrecMax	35.8	2.9	2.4	2.5	3.6	6.1	6.0	7.7	6.9	4.1	5.4	3.4	2.6
	PrecMin	18.1	0.5	0.2	0.4	0.2	0.4	0.7	0.2	0.1	0.6	0.6	0.0	0.7
	PrecStd	4.4	0.6	0.6	0.6	0.8	1.1	1.4	1.8	1.8	0.9	1.1	0.9	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.6	41.4	56.4	73.8	82.0	85.6	88.9	85.9	76.3	63.5	50.7	47.2
		MCWB	39.9	37.8	45.1	56.0	66.0	68.8	72.5	71.4	65.2	55.2	48.1	45.0
	2%	DB	38.4	39.1	49.2	67.3	76.2	81.1	84.5	80.9	70.7	58.1	47.8	43.4
		MCWB	36.7	36.5	40.8	52.5	61.3	67.3	70.1	67.4	61.2	53.1	45.3	41.5
	5%	DB	36.2	37.1	44.0	62.1	72.0	76.7	80.9	77.3	66.9	55.2	45.5	40.5
		MCWB	34.7	34.9	38.5	49.9	58.7	64.5	68.7	65.9	59.2	51.6	43.5	39.0
	10%	DB	34.8	35.5	40.4	57.3	68.0	73.3	77.5	73.7	63.6	52.9	43.2	38.0
		MCWB	33.5	33.8	36.5	47.6	56.6	62.4	66.9	64.4	57.6	49.9	41.5	36.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.4	38.8	46.0	57.7	68.9	71.6	75.0	72.7	66.0	56.6	48.7	45.1
		MCDB	41.3	40.4	55.2	70.1	80.0	82.8	85.9	82.9	74.3	60.6	50.0	46.8
	2%	WB	36.9	36.8	41.7	53.9	63.8	68.6	71.7	69.8	62.7	54.0	45.8	41.6
		MCDB	38.0	38.4	47.7	64.2	72.3	77.8	81.6	77.9	68.5	57.2	47.4	43.0
	5%	WB	34.9	35.3	39.2	51.3	60.9	66.4	69.9	67.4	60.5	52.3	43.7	39.2
		MCDB	35.9	36.8	43.0	60.2	68.5	74.0	78.7	74.3	65.3	55.0	45.2	40.4
	10%	WB	33.6	34.0	37.1	48.6	58.3	63.8	67.9	65.3	58.4	50.1	41.6	36.6
		MCDB	34.6	35.4	40.0	56.4	66.0	71.2	75.9	72.4	62.8	52.8	43.1	37.9
Mean Daily Temperature Range	5% DB	MDBR	8.8	11.1	13.2	17.0	18.6	17.1	17.1	17.4	15.2	10.2	7.1	7.5
		MCDBR	6.1	7.7	17.3	25.5	25.1	22.3	22.1	23.3	20.4	13.5	8.3	6.7
		MCWBR	5.8	6.2	11.1	13.3	12.6	10.8	10.5	11.5	11.8	9.6	7.5	6.3
	5% WB	MCDBR	5.9	6.5	15.0	22.1	20.7	19.4	20.0	19.8	17.9	11.8	7.9	6.5
		MCWBR	5.8	5.7	10.3	12.3	12.0	9.9	10.4	11.1	11.4	8.9	7.5	6.3
Clear-Sky Solar Irradiance	taub	0.284	0.322	0.343	0.386	0.376	0.372	0.398	0.399	0.365	0.343	0.320	0.277	
	taud	2.117	2.110	2.181	2.268	2.379	2.413	2.356	2.344	2.414	2.372	2.215	2.115	
	Ebn at Noon	170	217	251	260	272	275	264	254	245	214	156	134	
	Edh at Noon	17	27	33	36	35	34	35	33	27	21	16	12	
All-Sky Solar Radiation	RadAvg	120	345	789	1211	1656	1721	1662	1319	827	355	113	68	
	RadStd	26	62	107	152	157	162	184	115	92	64	18	15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	-284	-326	N/A	N/A

Nomenclature: See separate page